SUVADEEP MAITI

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Examination	Institute	Year	GPA
Ph.D. in Biomedical Engg & Medical Physics	University College London, London	2024 - Current	NA
M.S. in Electronics & Coomunication Engg	IIITH, Hyderabad	2021 - 24	8.57
B.E. in Electrical Engg	Jadavpur University, Kolkata	2016 - 20	8.93

RESEARCH INTERESTS

* Neurotechnology * Deep Learning * Deep Brain Stimulation * Wearable Devices

RESEARCH EXPERIENCE

- **University College London**
  *PhD student – [Advisor: **Prof. Vladimir Litvak**]* *(Oct'24-Present)*
  - **Wearable EEG for Radiation-Free DBS Lead Orientation**
    - Developing a radiation-free EEG-based method to measure segmented Deep Brain Stimulation (DBS) lead orientation, focusing on optimization for outpatient clinical use.
  - **spm-python: Python Bindings for SPM25**
    - developing spm-python, a Python interface for SPM that enables its use without a MATLAB license.
- **International Institute of Information Technology, Hyderabad (IIIT-H)**
  *Research Fellow – [Advisor: **Prof. Bapi Raju**]* *(Jan'22-Aug'24)*
  - **India Sleep Dataset**
    - in collaboration with NIMHANS Bangalore, developed a user-friendly preprocessing pipeline to detect sleep disorders (apnea, snoring, limb movements).
  - **Resource Constrained Brain Activity Based Sleep Stage Classification**
    - Concept Used: LSTM, Residual Networks, Attention & Squeeze-and-Excitation Block
    - Developed deep-learning technique for classifying sleep into 5 different stages as required by sleep clinicians.
  - **Eye movement based Sleep Tracking**
    - Concept Used: Transformer Encoder & Squeeze-and-Excitation Networks
    - Developed sleep staging algorithm utilizing EOG signals, working on a sleep-mask based EOG signal acquisition wearable device.
- **Institut des Sciences Cognitives Marc Jeannerod, CNRS, UMR 5229, France**
  *Research Intern – [Advisor: **Dr. James Bonaiuto**]* *(May'23-Aug'23)*
  Worked on laminar source reconstruction with high-precision MEG
  - **There's life in that old MEG yet: Depth electrode-like laminar source reconstruction with high precision MEG.**
    - Concept Used: Laminar MEG, Source Reconstruction Algorithms.
    - Implemented simulations to explore the current source density (CSD) transform as part of a project aiming to develop laminar source reconstruction for MEG data.

PUBLICATIONS

- **Polysomnography Dataset for Sleep Analysis in Ischemic Stroke Patients.**, *Suvadeep Maiti**, Shivam Kumar Sharma*, S. Mythirayee*, P R Srijithesh and Raju Surampudi Bapi. *Nature Scientific Data) 2026*, https://doi.org/10.1038/s41597-026-06747-w.

- **Explaining Inaccuracies in Patient EEG Based Sleep Staging**, *Suvadeep Maiti**, Saswata Bose, Shivam Kumar Sharma*, S. Mythirayee*, P R Srijithesh, Tapabrata Chakraborty and Raju Surampudi Bapi. *accepted at IEEE International Symposium on Biomedical Imaging (ISBI) 2026.*
- **Enhancing Healthcare with EOG: A Novel Approach to Sleep Stage Classification**, *Suvadeep Maiti**, Shivam Kumar Sharma*, Bapi Surampudi Raju. *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP) 2024, pp. 2305-2309, doi:10.1109/ICASSP48485.2024.10446703.*
- **Transparency in Sleep Staging: Deep Learning Method for EEG Sleep Stage Classification with Model Interpretability**, Shivam Kumar Sharma*, *Suvadeep Maiti**, S. Mythirayee, P R Srijithesh and Raju Surampudi Bapi. *Under review at Journal of Biomedical and Health Informatics*, arxiv.org/pdf/2309.07156.
- **There's life in that old MEG yet: Depth electrode-like laminar source reconstruction with high precision MEG.**, Maciek J Szul, *Suvadeep Maiti*, Ishita Agarwal, Siqi Zhang, Gareth R Barnes, Sven Bestmann, James J Bonaiuto. *MEG-UKI 2023*, meguk.ac.uk/meg-uki-2023-conference/poster-session-2/.

** indicates equal contribution* | *Full publication list at Google Scholar* | *Project details at Website*

SELECTED PROJECTS

- **Wilson-Cowan model to investigate Working Memory** | *Cog Sci Lab, IIIT-Hyderabad* *(Jan'22-May'22)*
  - Implemented Wilson-Cowan model to describe dynamics of interactions between populations of simple excitatory and inhibitory model neurons
- **RoadFinder: Mapping Roads with K-Means** | *Image Processing Course Project, IIIT-Hyderabad* *(Aug'21-Nov'21)*
  - Implemented the K-means algorithm on satellite images to isolate roads, applied morphological operations for road extraction, and compared results with a reference image for evaluation.
- **Building LeNet-5 from Scratch for Brain-Score Evaluation** | *SMAI Course Project, IIIT-Hyderabad* *(Aug'21-Nov'21)*
  - Designed a 7-layer LeNet-5 architecture, evaluated it on the MNIST dataset, and used the Brain-Score platform to benchmark its performance against standardized ventral stream computational models.

TECHNICAL SKILLS

Programming Language: Python, Matlab

Library: Scikit-learn, Numpy, Matplotlib, Pandas, Nilearn, Brian2

Framework: PyTorch, PyTorch Lightening

Techonologies & tools: draw.io, Anaconda (Python), LINUX, LaTeX, MS-excel

TEACHING

- Teaching Assistant for Neuromatch Academy Computational Neuroscience | *Neuromatch Academy* *(July'24)*
- Teaching Assistant for CS9.427.S22 Introduction to Neural and Cognitive Modeling | *IIIT-Hyderabad* *(Aug'22-Dec'22)*
  - Taught by Prof. Bapi S. Raju
- Teaching Assistant for CS9.423.S23 Cognitive Science and AI | *IIIT-Hyderabad* *(Jan'23-May'23)*
  - Taught by Prof. Bapi S. Raju

AWARDS & ACHIEVEMENTS

- Winner of Deep Learning Challenge at London Neurotech Hackathon *(Jan'25)*
- **UCL ISAD Tuition Fee Support Award** *University College London, Oct'24*
  Award value: £33,000 per year for 4 years (total £132,000)
- **UCL EPSRC DTP Research Studentship** *University College London, Jan'25*
  Award value: £22,800 per year for 4 years (total £91,200)
- Invited Guest Speaker at Dalian University of Technology *(Apr'24)*
- **CNRS Research Fellowship** *Centre National de la Recherche Scientifique, France (May'23–Aug'23)*
  Award value: €700 per month for 3 months (total €2,100)